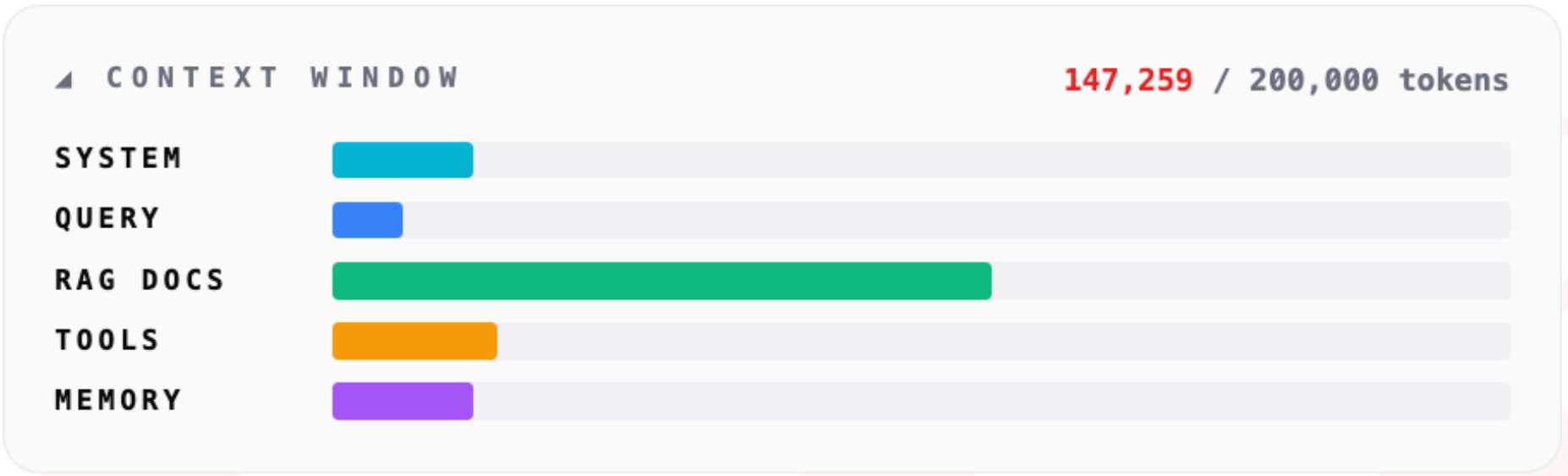
ART · SCIENCE · ENGINEERING

Context *engineering.*

The art and science of filling the context window — the missing skill behind every great AI output in 2026.



WHY THIS MATTERS

Ever wondered why AI sometimes feels *clueless?*

It isn't the prompt. It isn't the model. It's the **context** — everything you do (or don't) hand to the model before it starts thinking.

WITHOUT CONTEXT

Generic. Shallow. Confidently wrong.

WITH CONTEXT

Precise. Personal. Useful.

Welcome to **Context Engineering** — where better inputs build smarter outputs.



SYSTEM DESIGN INTERVIEW

When a celebrity tweets, millions of followers see it almost instantly.

How does Twitter fan out and deliver tweets at that scale?

ANSWER ↓ IN THE CAPTION



Akhila ✓
@akhila

FOLLOW +

DEFINITION

What is *context* engineering?

> The method of building the right environment around a prompt — so an LLM can answer with relevance, not guesswork.

IT'S NOT JUST PROMPTING — IT'S INPUT ARCHITECTURE.

Memory
What was said before.

User Data
Who they are, what they need.

Tone
How the model should speak.

Goals
What success looks like.

...so models like **Claude** & **GPT** can produce relevant, accurate, personalized answers.

SIDE BY SIDE

Prompt vs *Context* engineering.

PROMPT ENGINEERING

Writing the question.

- Tweak words, structure, examples.
- One-shot or few-shot framing.
- Works inside one message.
- Optimizes wording.

CONTEXT ENGINEERING

Building the environment.

- Curate what enters the context window.
- Pull memory, docs, tools, history.
- Operates across the entire session.
- Optimizes signal-to-noise.

Prompt is what you **say**. Context is everything the model **knows** when you say it.

TAXONOMY

The 4 main types of *context.*

01

User Context

Who is asking?

Preferences, role, history, goals.

02

Task Context

What's the job?

Write, analyze, refactor, decide.

03

Temporal Context

When is this happening?

Last message · this week · pre-deadline.

04

Environmental Context

Where is this used?

Mobile, CRM, support desk, IDE.

Each layer adds **depth** and **clarity** to what the model can do.

ELEMENTS

What actually fills the *context window.*

01 **System Prompt**
The role, tone, and rules the model must follow.

02 **User Query**
The actual question or task at hand.

03 **Retrieved Docs**
RAG hits, knowledge base snippets, citations.

04 **Tool Results**
Outputs from function calls, search, code execution.

05 **Memory**
Past turns, user profile, long-term notes.

Every token has a job. If it's not earning its place, it's stealing room from something that would.

PIPELINE

How models *read* the context.



STEP 01 **Input**

Prompt + retrieved context arrives.



STEP 02 **Tokenize**

Words split into model-friendly tokens.



STEP 03 **Attend**

Attention scores weight every token.



STEP 04 **Generate**

Next-token prediction, one at a time.

Better context → sharper attention → **better next token.**

IN PRACTICE

Same model. *Different context.*

Support Agent	USER HISTORY + ORDER CONTEXT
NO CONTEXT *"Hi, how can I help?"*	WITH CONTEXT **"Hey Priya, I see your last order #4821 hasn't shipped — let me check on that."**

Sales Copilot	ACCOUNT RESEARCH + RECENT NEWS
NO CONTEXT *"Here are 3 generic email templates."*	WITH CONTEXT **"Based on Acme's recent fundraise, here's a tailored opener for their CTO."**

Coding Assistant	REPO FILES + CODE CONVENTIONS
NO CONTEXT *"Here's how to write a React hook."*	WITH CONTEXT **"In this repo you use Zustand — here's a `useCart` hook that fits your store pattern."**

TOOLKIT

Build your *context stack.*

01 ORCHESTRATION

LangChain

Chain prompts, tools, retrievers.

02 RAG

LlamaIndex

Indexing, chunking, retrieval.

03 VECTOR DB

Pinecone

Fast semantic search at scale.

04 MEMORY

Mem0

Long-term agent memory store.

05 TOOLS

Claude · MCP

Plug live tools into the context.

06 IDE

Cursor

Repo-aware code context.

Pick one for **retrieval**, one for **memory**, one for **tools** — that's a stack.

BETTER CONTEXT · BETTER AI

Stop tweaking prompts. *Start engineering context.*

The prompt is only the tip. The real lift is in what surrounds it — memory, retrieval, tools, and intent. That's where the next decade of AI work lives.

THREE THINGS TO DO TODAY

01 Audit your prompts — what context are they missing?

02 Wire in retrieval before you tweak the wording.

03 Save what you learn into long-term memory.

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